

Flood Risk Analysis for New York City Using predictive Modeling

1. Introduction

Flood risk poses a significant challenge in urban areas, especially densely populated regions like New York City. Urban flooding results from inadequate drainage systems, limited storage capacity, insufficient maintenance, and unplanned urban growth (Eini et al., 2020). These risks are exacerbated by climate change and land use changes. Flood risk depends on both flood hazards and vulnerability. In this case study, I conduct a thorough flood risk analysis to identify vulnerable zones within New York City, aiming to optimize evacuation processes and assess potential flood risk.

This study examines flood vulnerability by considering factors like elevation, rainfall, and slope characteristics. Additionally, it assesses distinctive urban risk elements, such as the built environment (including commercial and residential structures, hospitals, schools, universities, and government buildings), emergency facilities (police stations and fire stations), and demographic data at the Neighborhood Tabulation Area (NTA) level in New York City. Furthermore, the report investigates ways to enhance evacuation processes using geospatial data and addresses ethical considerations in decision-making.

2. Methodology

2.1 Data Collection and Integration

In this study, I employed a range of geospatial data, including both raster and vector formats. My approach aligns with the methodology used by Eini et al. (2020), where they defined Flood risk as the likelihood of loss resulting from flooding and depends on elements susceptible to it. It is calculated by multiplying the probability of flood by the potential adverse consequences for human life and the urban environment:

$$\text{Risk} = \text{Hazard} \times \text{Vulnerability} \quad 1)$$

Flood vulnerability is defined as the degree of susceptibility to damage in an area or population and its ability to resist. In other words, against an identical flood, the amount of damage depends on the area's characteristics and demographics.

The following data sources were integrated to create Flood Risk map and identify vulnerable zones:

- **Elevation Data:** High-resolution Digital Elevation Models (DEMs) for New York City(1 Foot Digital Elevation Model (DEM), 2013).
- **Rainfall Data:** 5 year Historical rainfall data over New York City was used (2015-2020). (Giovanni, n.d.)
- **Slope:** Gradient of the topography, which helps to identify locations with flat or steep gradients.
- **Hydrological/Water Data:** Distance to shoreline, and water bodies. (NYC Planimetric Database: Shoreline, n.d.)
- **Infrastructure Data:** Critical infrastructure such as roads, bridges, hospitals, government, and residential zones. (NYC Planning Zola, n.d.)
- **Population Data:** Demographic information to identify vulnerable populations.(NYC Street Map, n.d.), (CoreData.Nyc, n.d.) (Census, n.d.)

2.2 Predictive Modeling

Flood Hazard Model: I employed the Genetic Algorithm for Rule-set Prediction (GARP) to create a flood hazard map based on environmental variables (Elevation, Rainfall, Slope). The genetic algorithm evolves a population of rule sets to find the best set of rules for predicting flood occurrences based on environmental variables. The fitness function evaluates these rules, and the best individual is used to generate a flood hazard map. This approach is effective for initial flood risk assessment and can be refined with actual flood data for more accurate predictions in the future. The evaluation of the model wasn't carried out due to lack of historical flood data.

2.3 Vulnerability Assessment

Vulnerability is defined as factors social and economic characteristics that are sensitive to flood risk. I estimated vulnerability based on specific neighborhoods within NYC, by looking at their key characteristics, such as state of infrastructure, availability of emergency services, susceptibility to water bodies, and shorelines.

Vulnerability	Sub criteria
Infrastructure	Street, hospital, school, building quality
Water	Distance to shoreline, FEMA flood map zone, distance to water bodies
Emergency	Density of emergency services

Each factor influencing flood risk, such as infrastructure, emergency services, and proximity to water bodies, was evaluated using equation 2 (Eini et al., 2020). The vulnerability index was estimated by measuring the density of each sub-criterion within each unique neighborhood tabulation area. For example, infrastructure, a critical factor in flood impact, was assessed

based on its quality and the ratio of commercial to residential buildings. Additionally, the quality of buildings was evaluated by estimating their age and recent renovation cycle (*BYTES of the BIG APPLE - DCP*, n.d.). This information was aggregated at the NTA level and analyzed using the vulnerability equation. Ultimately, these factors were weighted and assessed to create an overall infrastructure rating score.

Vulnerability estimation: Ensure each vulnerability sub-criterion (economic, social, infrastructural) is normalized. This means scaling the values between 0 and 1, where 0 represents the least vulnerable and 1 represents the most vulnerable.

Example normalization formula:

$$V_r^{std} = \frac{V_r - V_{min}}{V_{max} - V_{min}}$$

2)

where V_{std} is the normalized vulnerability, V_r is the raw value, V_{min} and V_{max} are the minimum and maximum values for that criterion.

Calculate Weighted Vulnerability

1. **Weight Assignment:** I employed methods such as Analytical Hierarchical Process (AHP) and Principal component analysis (PCA) to assign weights to each sub-criterion based on its importance.
2. **Overall Vulnerability Map:** Combine the weighted sub-criteria to create an overall vulnerability map.

Infrastructure Vulnerability

Infrastructure vulnerability was assessed using the following sub-criteria:

- **Urban Density:** The concentration of buildings in an area. Higher density implies more buildings at risk during flooding.
- **Quality of Buildings:** The structural integrity of buildings. Areas with lower quality buildings are more susceptible to damage.

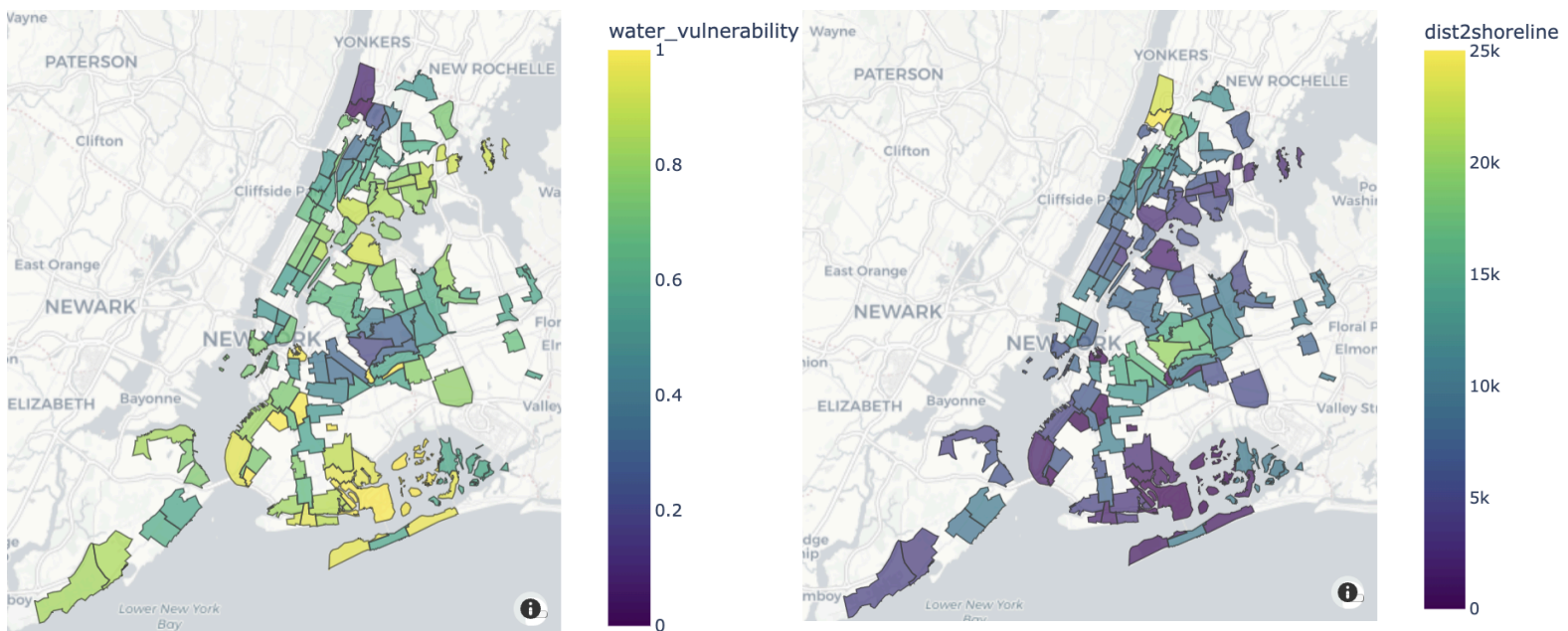
Infrastructure vulnerability, interestingly the map shows areas near the shoreline are at medium level risk, whereas areas farther from the source are lowly affected.

Quality of building can be a factor in assessing a properties ability to resist any natural catastrophe, areas with higher quality means that they are maintained in better condition than others.

Water Vulnerability

Water vulnerability was assessed based on:

- **Proximity to Water bodies and shoreline:** average distance to water bodies was evaluated at NTA level, neighborhoods closer to these shoreline are more vulnerable during floods



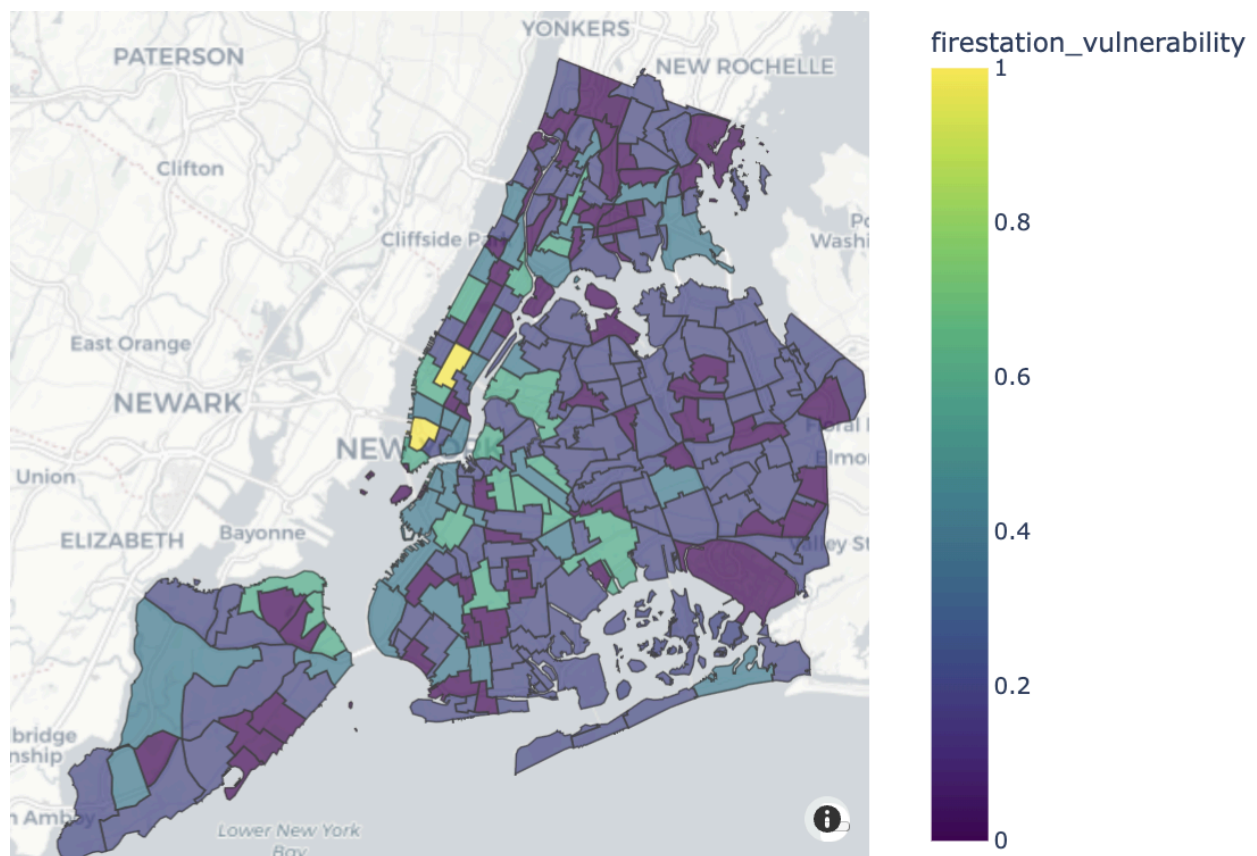
Emergency Vulnerability

- **Availability of emergency services:** NTA level concentration of emergency services that can provide resistance or aid in evacuation in case of a flood.

The figure above shows direct correlation to vulnerability based on its average distance to Shoreline. Additionally, I also evaluated FEMA insurance premium data for this evaluation.

Emergency Vulnerability

The emergency vulnerability index was measured by estimating the density of fire stations and other ancillary emergency service stations available in NYC. From the map, we can observe that Lower Manhattan has a higher concentration of these services compared to neighborhoods farther away.



Social Vulnerability

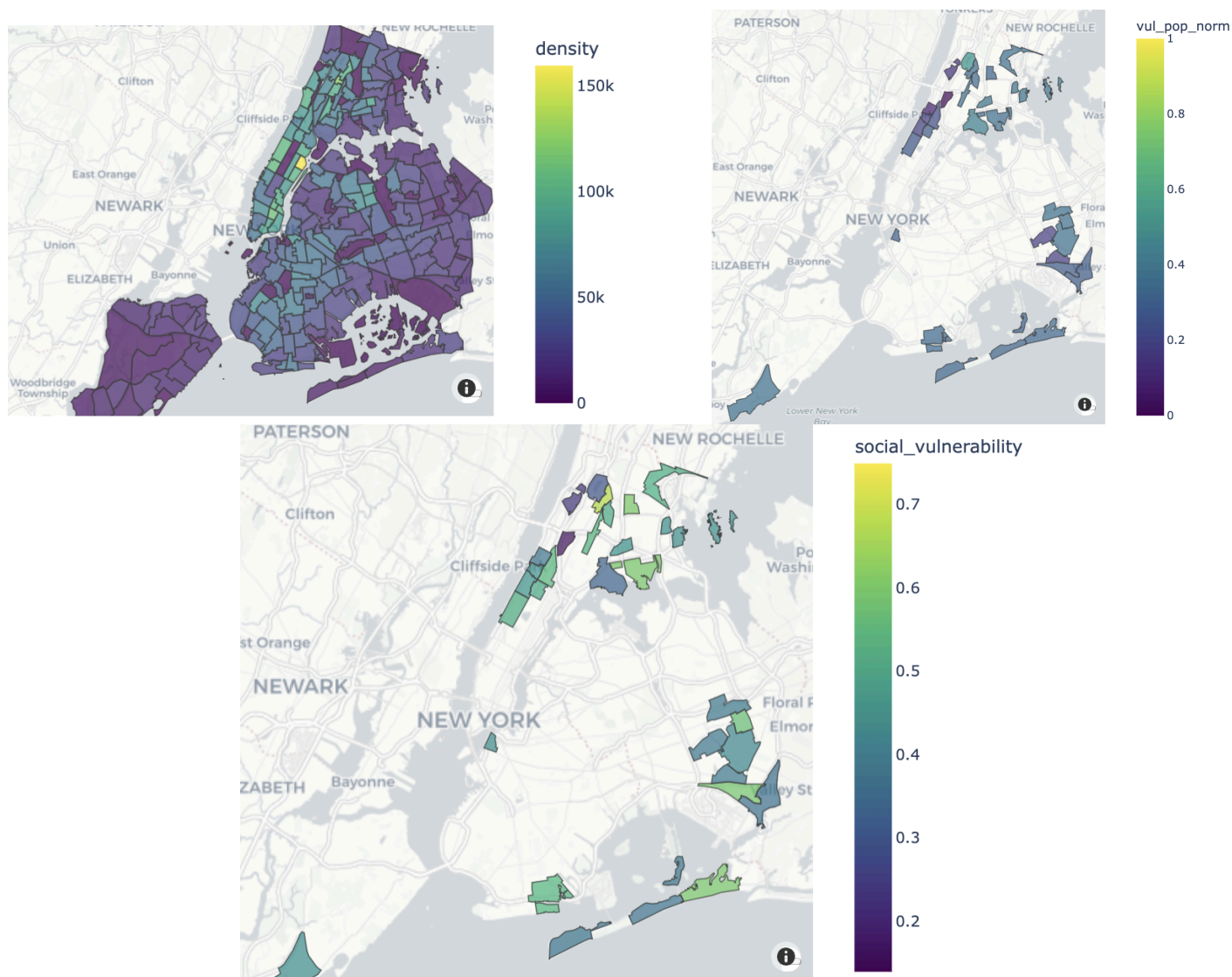
Social vulnerability was evaluated using the demography data, but wasn't eventually used in the final model development due to key missing data at NTA level:

I evaluated demography in the sub categories described below.

- **Age Structure:** Population under 20 and over 65 years old, who are more vulnerable due to physical and decision-making limitations.
- **Population Density:** Higher population density areas are more vulnerable.
- **Ethnic Diversity:** Neighborhoods with diverse ethnicities face communication and coordination challenges during crises.

These sub-criteria were classified and normalized similarly to the economic vulnerability sub-criteria.

The figure above shows the concentration of vulnerable populations, such as individuals over 65 and under 18 years old. The data for this is sparse and not available for each NTA. However, it can be observed that areas in Queens and the upper region of Manhattan have a high concentration of vulnerable populations, necessitating immediate evacuation.



The plot above illustrates the derived social vulnerability index based on demographic data. Although incomplete, it suggests the range of high vulnerability in NYC. Several key neighborhoods near the shoreline exhibit medium social vulnerability. Due to sparse data, this index was not included in the final vulnerability calculation; however, it should be considered in real models.

2.4 Combining Hazard and Vulnerability Maps

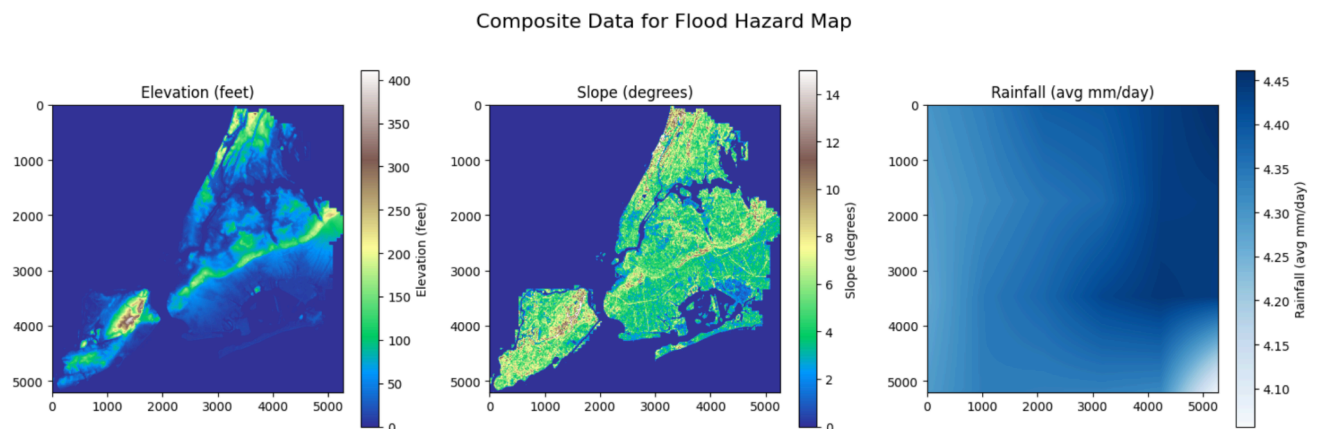
The overall flood risk map was produced by combining the flood hazard map with the flood vulnerability map using a weighted linear combination method. The weights for each criterion were determined using the Analytical Hierarchical Process (AHP). The overall procedure involved combining Hazard map (raster) derived from Elevation, slope and rainfall characteristics and Vector data which are each unique socio economic indexes evaluated specifically for NYC based on the current tabular data at NTA level.

3. Flood Models

3.1 Elevation Model

The elevation model illustrates the topography of New York City, highlighting low-lying areas that are more susceptible to flooding.

The rainfall model presents historical precipitation data, identifying regions with higher rainfall intensity and frequency.

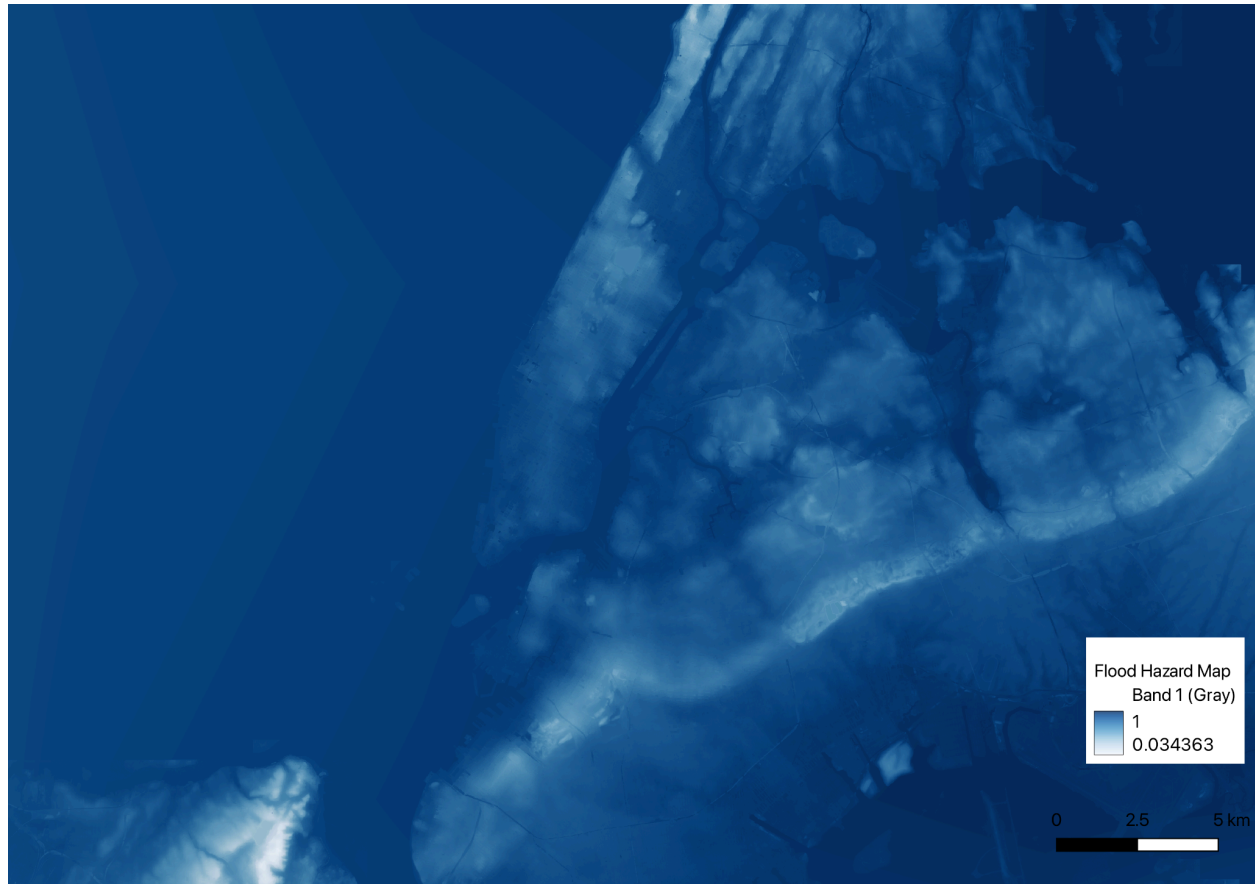


The slope model shows the gradient of the terrain, which affects water flow and accumulation during heavy rainfall.

3.2 Combined Flood Hazard Model

The Flood Hazard model integrates elevation, rainfall, and slope data to provide a comprehensive assessment of flood risk. The hazard model highlights areas of high risk, highlighted by areas with darker shaded hues, contrasting with areas of low flood hazard probability. The model generally follows Elevation terrain trends, indicating that higher elevated

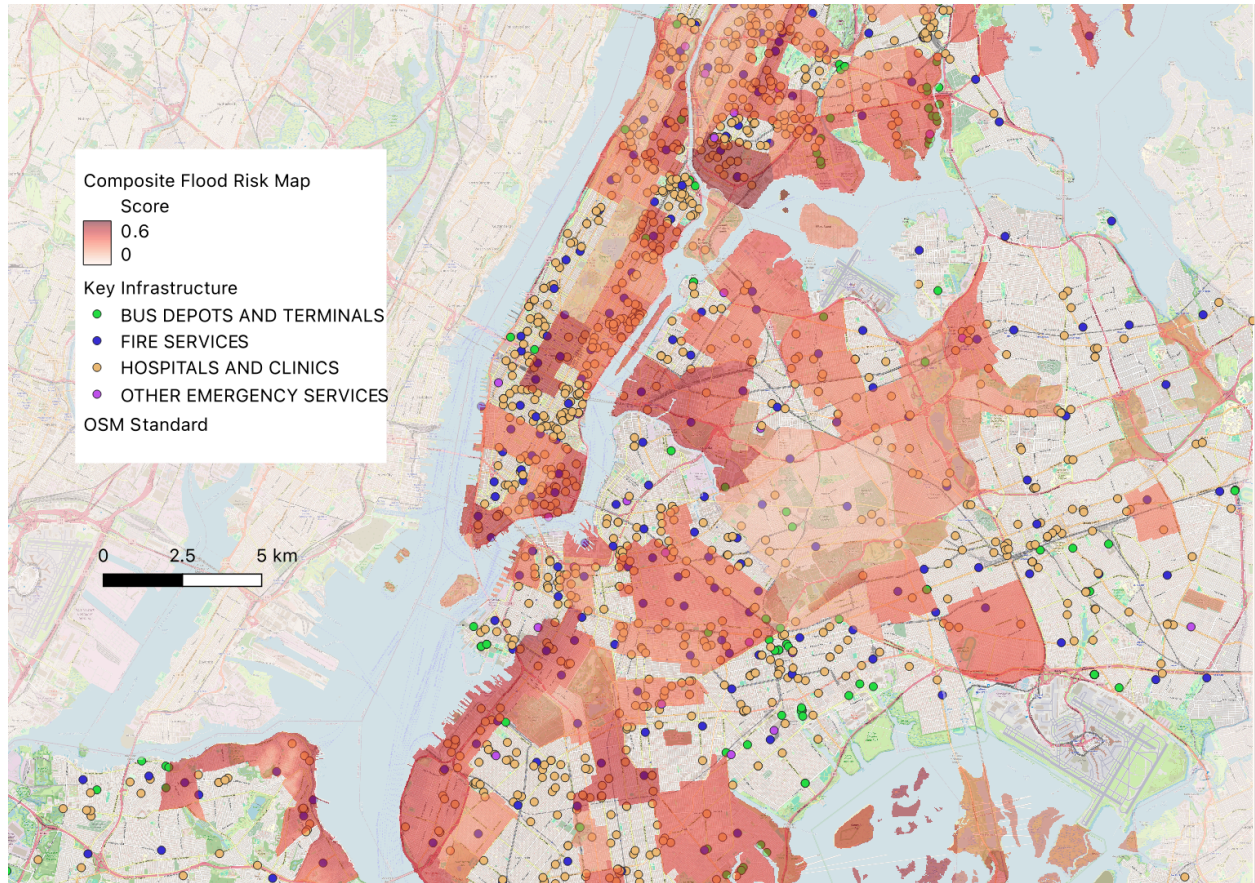
regions are less susceptible to flooding, whereas areas near shorelines and lower elevated regions are more prone to floods.



Composite hazard map, Elevation, Rainfall and Slope

3.3 Composite Flood Risk Model

A composite flood risk model was prepared by applying equation 1. First, a combined vulnerability score was created by assessing all risks posed by each unique vulnerability factor (infrastructure, water, and emergency). Since the vulnerability values were evaluated in vector format at the NTA level, each geometry was converted into raster format to match the dimensions of the Flood hazard model. For each unique grid point, the flood hazard value was multiplied by the vulnerability factor, resulting in a composite flood risk model. NTAs with missing data were not estimated, which is why the map shows missing NTA information. Additionally, sparse social demography data was not used. The results indicate that areas in Brooklyn, Lower Manhattan, and Queens near the shoreline are most sensitive to flood inundation and would require immediate emergency services.



4. Results

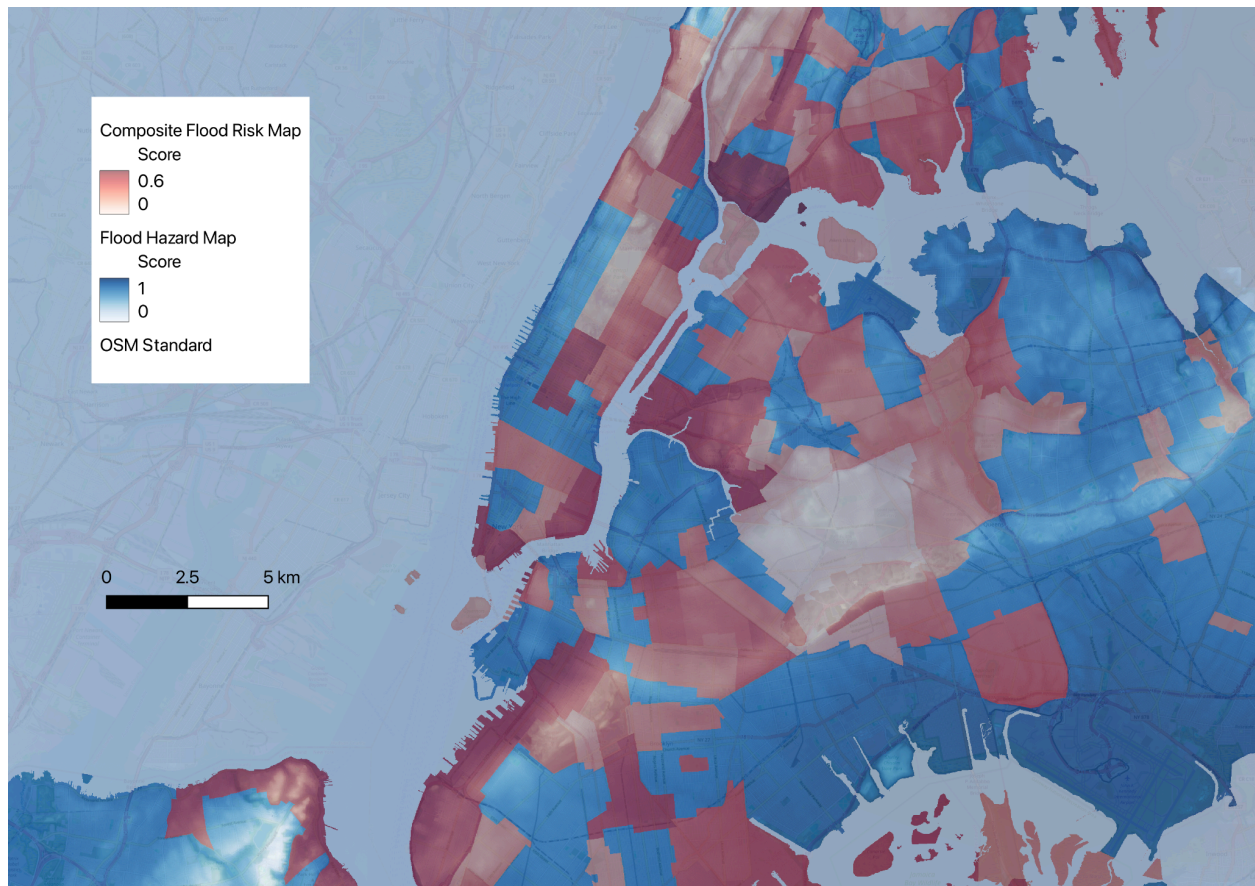
4.1 Flood Risk by NTA

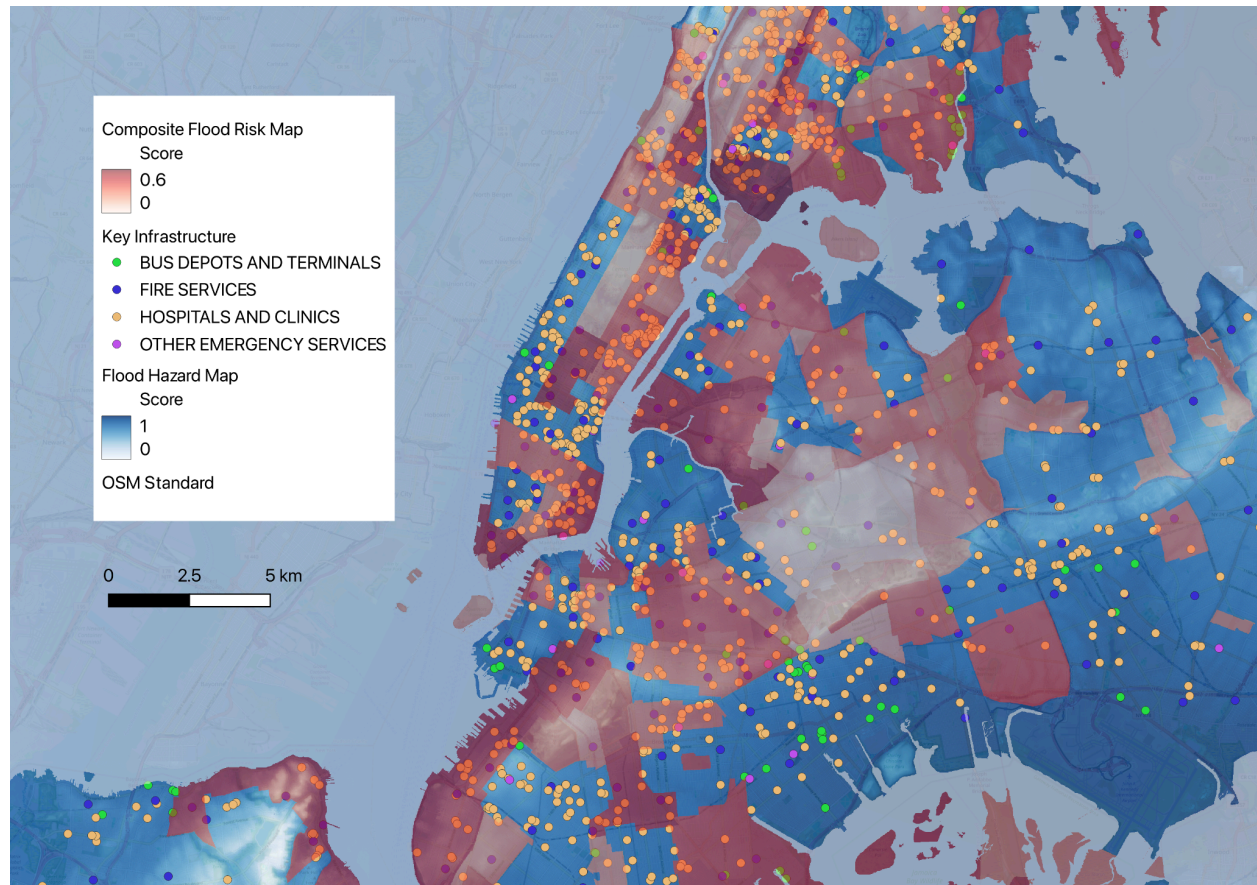
This section presents the flood risk assessment for each Neighborhood Tabulation Area (NTA), highlighting areas with high, medium, and low flood risk.

NTA Code	NTA Name	Borough	Population	Score
QN25	Corona	Queens	54870	0.28
BK27	Bath Beach	Brooklyn	29251	0.27
BK46	Ocean Parkway	Brooklyn	20427	0.24
QN26	North Corona	Queens	49596	0.18
BK88	Borough Park	Brooklyn	103706	0.16

4.2 Flood Risk Maps

Visual representations of flood risk across different NTAs.





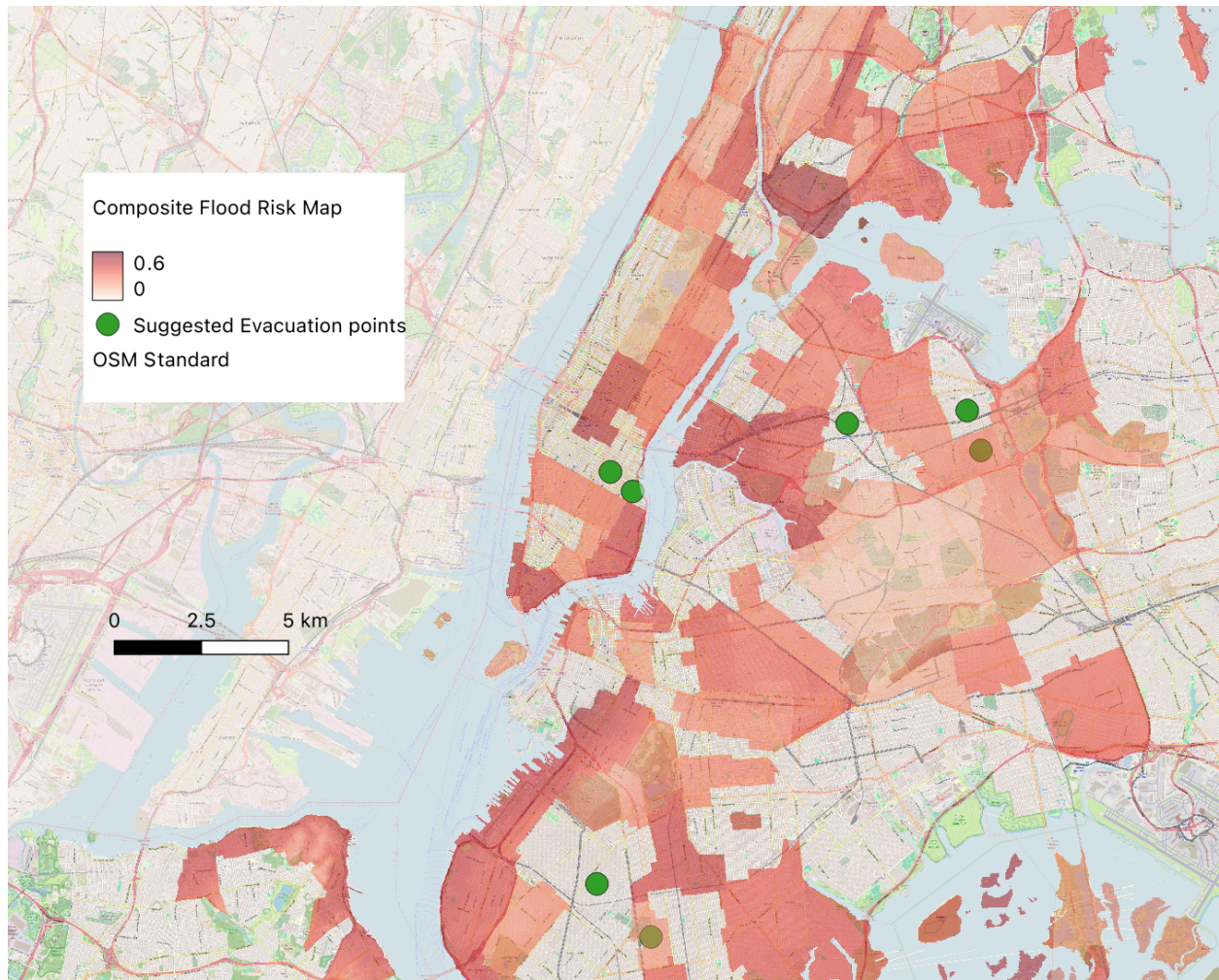
5. Discussion

5.1 Designing Geospatial Data for Evacuation Optimization

The flood risk maps, which considered factors such as infrastructure vulnerability, emergency response capacity, and shoreline sensitivity, revealed that Brooklyn, Queens, and Lower Manhattan had the highest neighborhood-level intensity scores. These areas, identified as critical for evacuation, have a high concentration of emergency services, allowing for immediate evacuation planning. Using these maps, a network-based geospatial analysis can be employed to determine the most efficient evacuation routes. Jin (n.d.) used shortest-path algorithms weighted by travel time rather than distance to optimize route planning during flood evacuations. Insani et al. (2022a) provided a comprehensive and adaptive solution for flood evacuation planning in densely populated urban areas, considering the dynamic and uncertain nature of demand and the critical time constraints imposed by impending floodwaters.

Clustering methods based on the density of bus depots, emergency services, and population can be used to assess evacuation points in the city. I evaluated potential evacuation spots

based on the centroid locations of NTA neighborhoods with the highest composite scores. The key locations are near the neighborhoods most prone to flooding. Additionally, the analysis identified other neighborhoods much farther from the NTA cluster, suggesting the presence of incomplete data.



5.2 Resource Allocation

- **Buses and Medical Support:** Locate resources outside the predicted impact zones based on flood risk assessments.
- **Shelter Locations:** Ensure emergency shelters are situated in safe areas with easy access for the affected population.

5.3 Ethical Considerations

- **Prioritizing Populations:** Address the ethical issues of prioritizing specific populations. Ensure decisions are based on vulnerability and need.

- **Incomplete Information:** Acknowledge the challenges of making decisions with incomplete information. Use the best available data and update plans as new information becomes available.
 - **Transparency and Fairness:** Ensure transparency and fairness in the evacuation plan. Communicate clearly with the public about the basis for decisions and the rationale for prioritizing certain areas or populations.
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6. Conclusions

The flood risk analysis for New York City using the GARP model provides valuable insights into the vulnerability of different neighborhoods to flooding. By integrating elevation, rainfall, and slope data, the study offers a detailed assessment that can aid in urban planning and disaster management. Future work can extend this analysis by incorporating additional environmental factors and exploring different modeling techniques.

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